DIP CHAKRABORTY

Software Engineer

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Backend Architecture · AI Integration · Enterprise Systems · IoT Data Pipelines · Technical Leadership

Software Engineer with 4+ years building AI-driven, enterprise-scale systems and leading cross-functional teams to deliver high-impact products for global clients.

PROFESSIONAL EXPERIENCE

Software Engineer

Preneur Lab Digital

January 2023 – Present Hybrid

- Led architecture and delivery of an AI-driven support and automation ecosystem (RAG, voice/chat, commerce workflows), scaling to **10k+ users/day**, reducing manual support effort by **90%** and SME admin overhead by **40%+**
- Architected cloud-native payroll and deployment infrastructure on GCP (Docker, CI/CD, monitoring), processing **5k+ employees/month** while sustaining **99.9% uptime** and cutting HR operational time by **25%**

Software Engineer

AIMS Lab

August 2021 – Dec 2022 Dhaka

- Engineered API middleware and C++/ROS robotic control systems for humanoid platforms, integrating real-time perception and task execution pipelines, improving system stability by **35%** and reducing testing time by **30%**

Software Developer

IftiSoft Ltd

March 2019 – July 2021 Dhaka

- Productized a WebRTC-based video conferencing solution using **Laravel + Vue.js** by conducting market validation and scalable prototyping, reducing development iteration cycles by **30%** and accelerating go-to-market readiness by **25%**

TOP PROJECTS

IoT Powered Access Control Solution

- Built an RFID-based access and identity platform with real-time device sync, supporting **5000+ daily users**
- Implemented data pipelines and digital wallet, eliminating manual reconciliation and reducing operational costs by **40%**
- Developed centralized analytics and predictive dashboards to optimize workforce utilization and facility operations

AI Powered Customer Support Agent

- Implemented RAG-based knowledge retrieval over **100k+ documents**, auto-resolving **65%+** queries and reducing repeat requests by **50%**
- Built voice, video, and chatbot automation handling **10k+ interactions/day**, enabling self-service workflows and reducing handling time from **6 min to <2 min**
- Developed sentiment-aware escalation and recommendation engines, cutting human support workload by **90%** and improving customer satisfaction by **30%+**

RESEARCH PROJECT

Autonomous Navigation for Mars Rover

- Co-led a cross-functional robotics team to design and qualify for the **World Final**, managing project budgets, recruitment, escalation, supply chain and technical reviews.
- Designed and implemented a GPS-denied autonomous navigation system using 3D mapping, visual odometry, and ROS-based planners in C++, achieving **95%+ waypoint accuracy** in field trials

TOP ACHIEVEMENT

- NASA Cansat Competition'24: Finalist
- University Rover Challenge'22: Finalist
- Google Hashcode'19: Regional Final

ACADEMICS

BSc, CSE

United International University

October 2017 – 2022

VOLUNTEERING

UIU Robotics

President

2020 – 2022 Dhaka, Bangladesh

- Led a **100+ member** robotics team, building technical training programs and mentoring project leads, improving competition performance by **30%+**
- Managed national competitions and major projects with **5M+ BDT** budgets, overseeing technical decisions and institutional reporting

RECOMMENDATIONS

Akib Zaman

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PhD Researcher, MIT

Ex-Project Director, UIU Mars Rover